

Assignment 3:

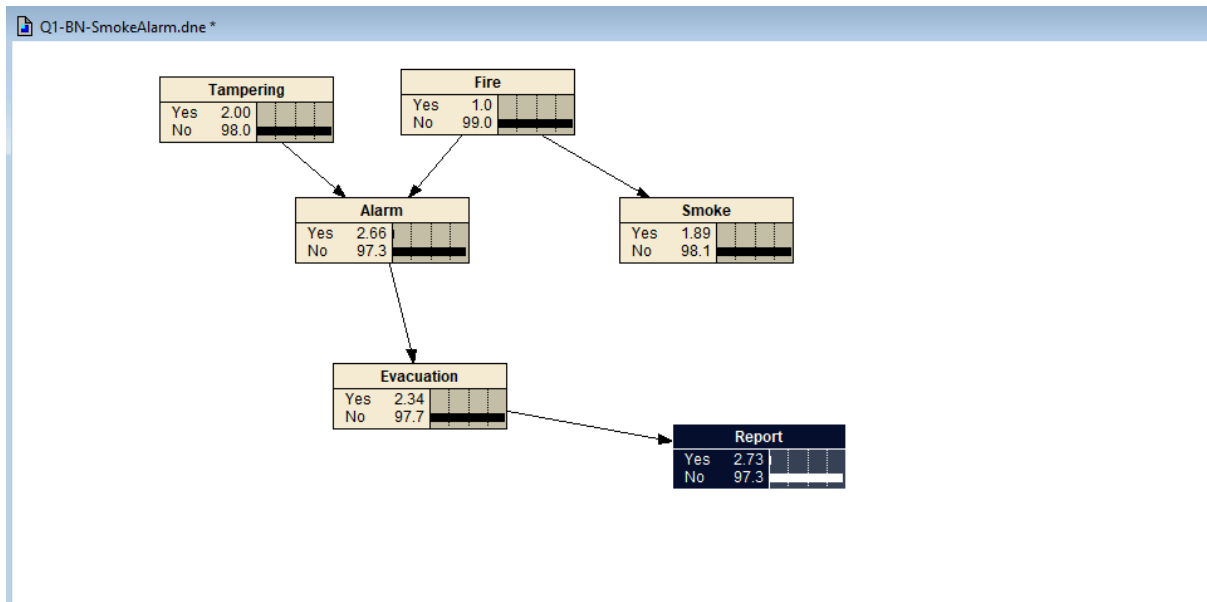
Bayesian Networks

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Question 1

(A) Extend your Bayesian Network by adding the appropriate nodes and edges. Provide the corresponding conditional probability tables (CPTs) for the new structure, and explain the reasoning behind your network design and CPT values.

Extended Bayesian Network:



Smoke CPT:

Smoke Table (in Bayes net SmokeAlarmBN)

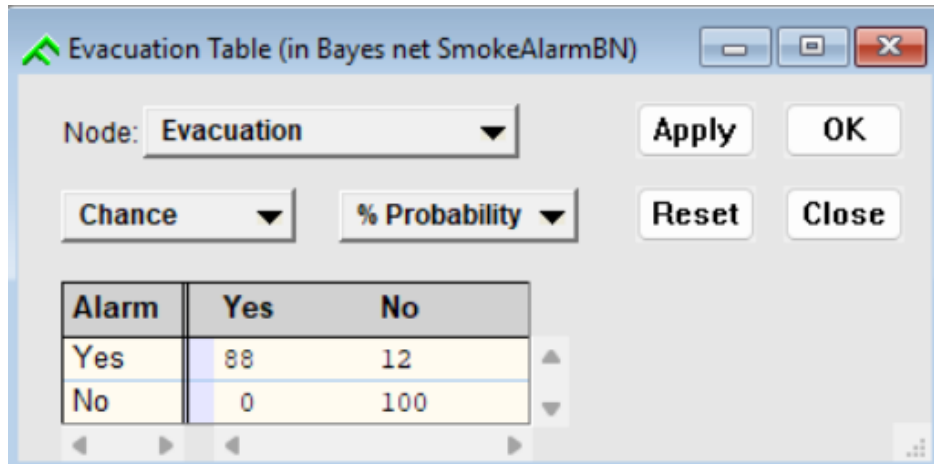
Node: **Smoke**

Chance % Probability

Reset Close

Fire	Yes	No
Yes	90	10
No	1	99

Evacuation CPT:



Evacuation Table (in Bayes net SmokeAlarmBN)

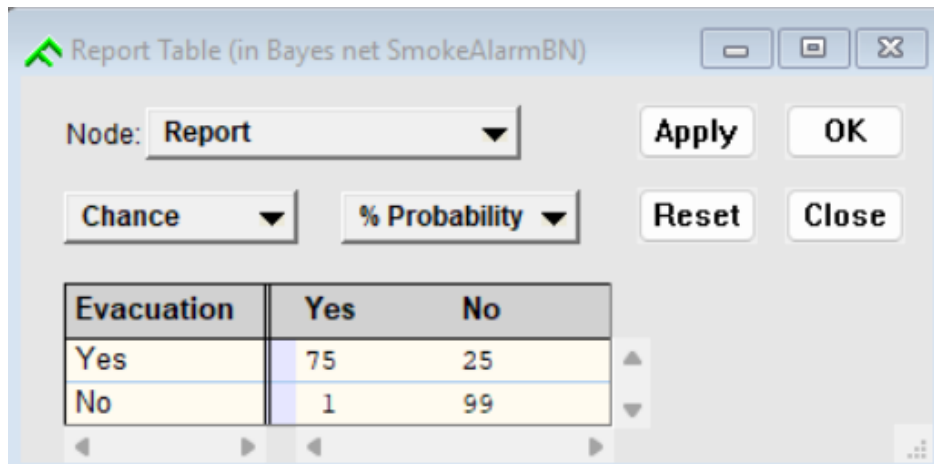
Node: **Evacuation**

Chance % Probability

Apply OK Reset Close

Alarm	Yes	No
Yes	88	12
No	0	100

Report CPT:



Report Table (in Bayes net SmokeAlarmBN)

Node: **Report**

Chance % Probability

Apply OK Reset Close

Evacuation	Yes	No
Yes	75	25
No	1	99

Justification - As stated in the assignment:

- Smoke is only produced when there is a fire, and hence, the smoke node is placed with fire as the parent node. As the probability of smoke when there is fire is 0.9 and the probability of smoke when there is no fire is 0.01. We take those values and fill in the CPT table.
- Evacuation only happens when there is a fire alarm, so the evacuation node is placed with the Fire Alarm as the parent node. As stated in the assignment, there is a 0.88 probability that you will have to evacuate when the alarm goes off, and no evacuation when there is no alarm.
- The newspaper only makes a report when there is an evacuation; hence, the report node has evacuation as its parent node. The CPT table shows that if there is an evacuation, there is a 0.75 probability that the newspaper will report it. If there is no evacuation, there is a 0.99 probability that the newspaper won't report it.

(b) Use Netica on the expanded BN to answer the following questions:

1. What is the marginal probability that your smoke detector has been tampered with?

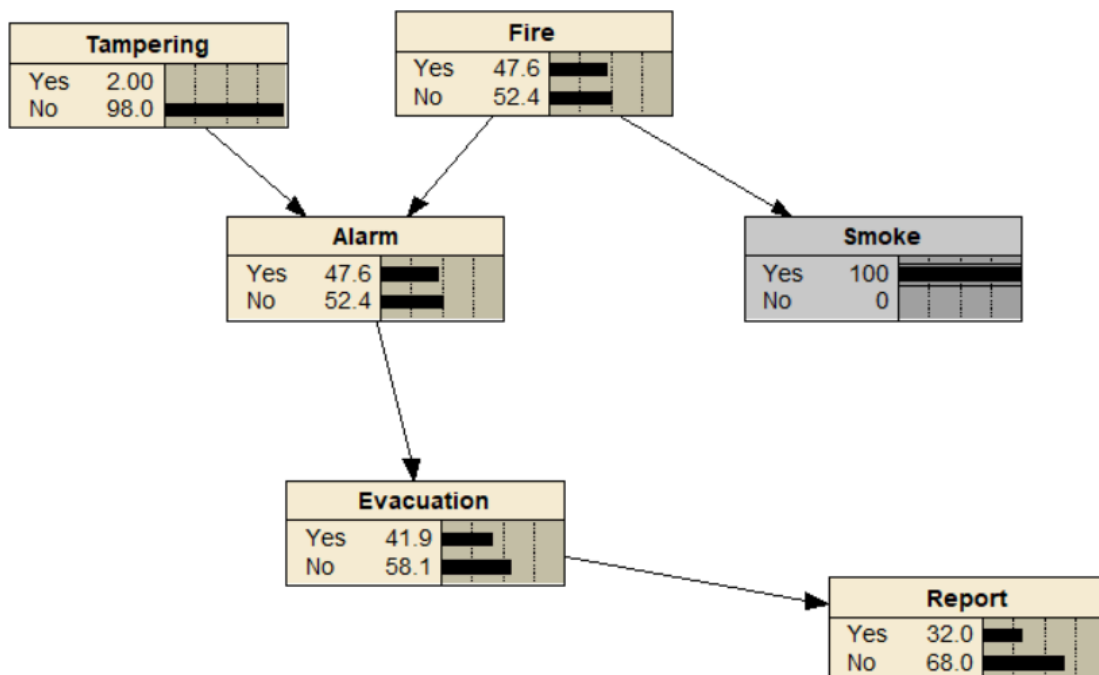
As there is no evidence in the following observation, we can conclude from the existing CPT that the marginal probability of our smoke detector being tampered with is 0.02

2. What is the marginal probability that there will be a news report tomorrow?

As there is no evidence that we have to act upon, we can see that the marginal probability of there being a news report tomorrow is 0.0273.

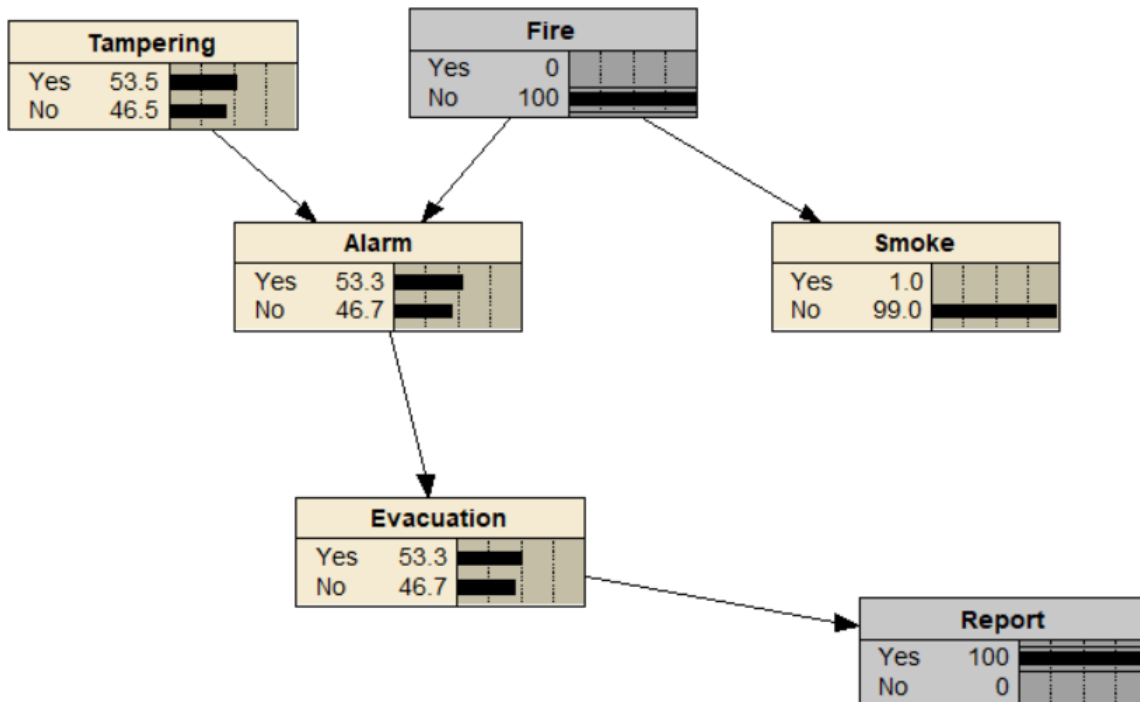
3. You have observed smoke in your apartment. What is the posterior probability that there will be a news report tomorrow?

Now using the evidence that we have observed smoke in the apartment, this helps us change the finding in the netica to set smoke to TRUE and upon changing that we see increase in the probability of fire which in turn increased the probability of evacuation and this also increases the probability of being a newspaper report which comes out to be 0.32. Therefore, we notice that if we have observed smoke in the apartment, there is a higher chance of a fire, which suggests the alarm being called off and evacuation happening.



4. You have observed no fire, and saw a news report about your apartment. What is the posterior probability that your smoke detector has been tampered with?

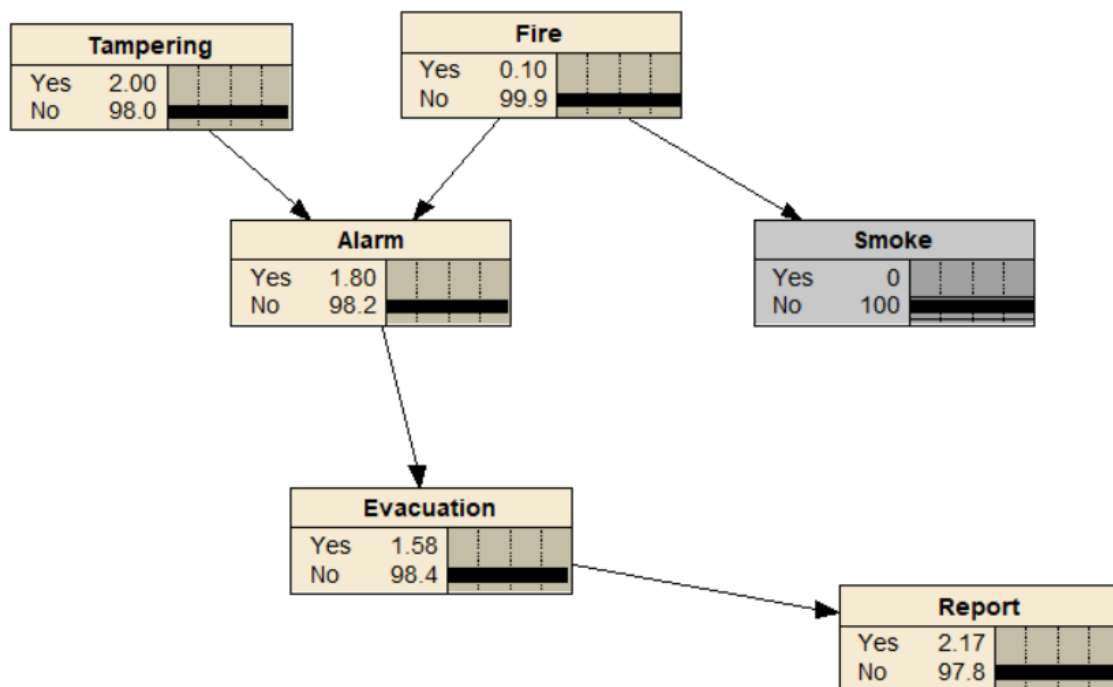
The evidence suggests no fire was observed, and there was a news report about the apartment. This suggests a higher chance that the alarm was tampered with, which is a plausible cause of the evacuation and, in turn, the report. The posterior probability of the tampering of the alarm is 0.535.



5. You have observed no smoke in your apartment. What is the posterior probability that your smoke detector has been tampered with? What conditional independence property could help you here?

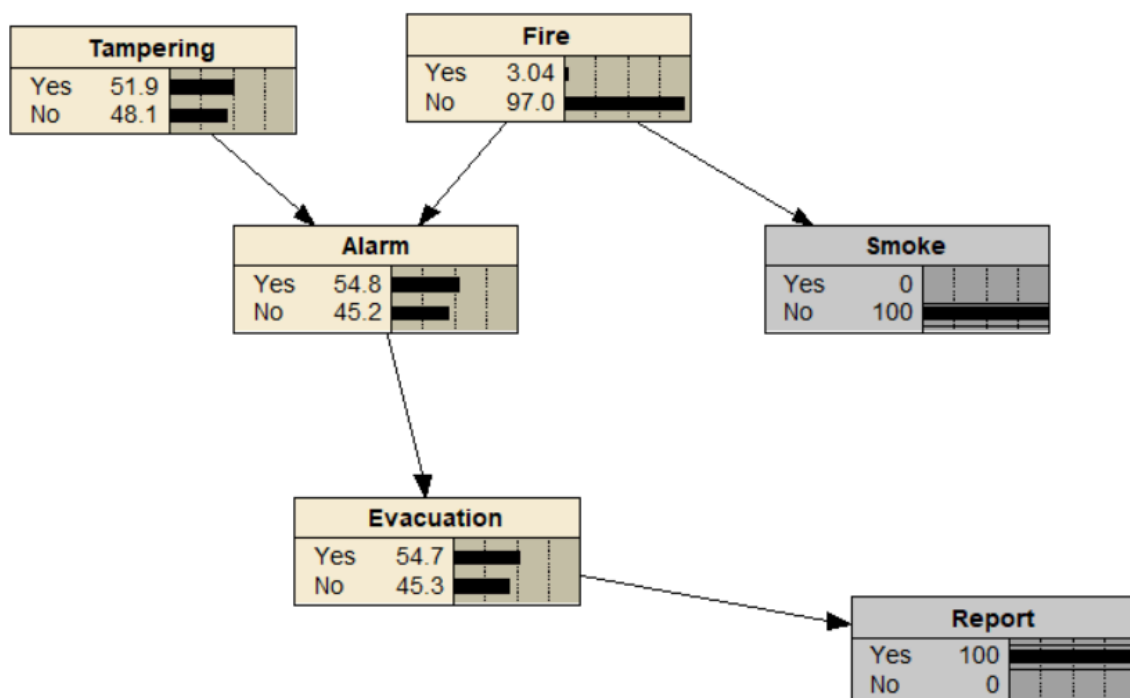
If no smoke has been detected, then the posterior probability of the smoke alarm being tampered with is 0.02. As smoke is only caused by fire, and the observation is that there is no smoke, hence the probability of fire is lower. But as tampering is not linked to smoke, the probability of there being tampering remains virtually unchanged.

This is an application of D-Separation using a common cause structure, as smoke and tampering are conditionally independent of each other under no given evidence, and there is no active path from tampering to smoke. Hence: Tampering $\perp$ with Smoke.



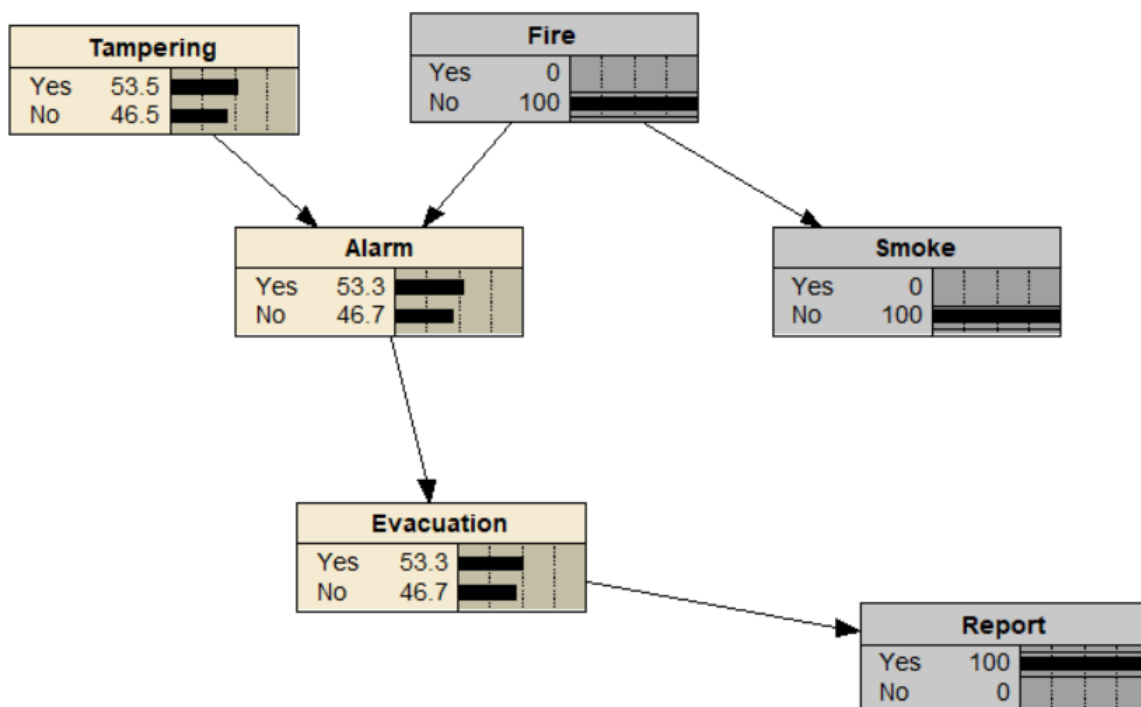
6. You have observed a news report about your apartment, and there is no smoke in your apartment. What is the posterior probability that your smoke detector has been tampered with? Given that the news report was observed, why does observing the absence of smoke affect your belief of whether or not your smoke alarm was tampered with?

Here we see that there was a news report about the apartment where no smoke was observed within the apartment. Now this raises the belief that the alarm was tampered with, therefore, we have a rise in the posterior probability of there being a tampering to 0.519, which is a significant increase. This mainly happens because a news report confirms that an evacuation was occurred which in turn signifies that the alarm was likely to be triggered but if there was no smoke and no fire then the most plausible explanation is tampering with the alarm, this is the most expected outcome due to common effect structure where outcome is based on the dependency between the causes.



7. You have observed no fire, saw a news report about your apartment, and there is smoke in your apartment. What is the posterior probability that your smoke detector has been tampered with? How does observing whether or not there is smoke affect your belief of whether or not your smoke detector has been tampered with? Why?

In the situation where a fire had not been observed in the apartment, and yet there was smoke in the apartment, a newspaper report was made. The posterior probability of there being a tampering of the smoke alarm is 0.535. This means that there was not a fire and yet a report was observed which implies that an evacuation must have happened and the alarm likely was triggered, but as there being smoke typically implies fire which was ruled out before tampering remains as the only plausible cause of there being a false smoke detection, hence the increase in tampering. Observing whether or not there is smoke by itself does not affect the belief in tampering, as they are not directly linked and are conditionally independent, but along with other observations about a report being released and evacuation happening or not, we can see a change in the observation of tampering.



(c) Hypothesise the (conditional) independence properties of the statements below. Use Netica to check them, and state whether they are true or false. Briefly justify your answers by indicating whether the variables in question are connected or D-separated, and which variable(s) block or unblock the path — use the terms of Common Cause, Common Effect and Causal Chain where applicable.

1. Tampering $\perp$ Evacuation

Path: Tampering $\rightarrow$ Alarm $\rightarrow$ Evacuation

Conditioning: None

As there is a direct path between Tampering Alarm and Evacuation, this causal chain is unblocked, which means that tampering and evacuation are connected. Therefore, tampering is not conditionally independent of Evacuation, and they are not D-separated.

2. Tampering $\perp$ Evacuation | Alarm

Path: Tampering $\rightarrow$ Alarm $\rightarrow$ Evacuation

Conditioning: Alarm is observed

As there is a middle node on this causal chain between tampering and evacuation, the path becomes blocked. Since alarm is the middle variable and we condition on it, the path from Tampering to Evacuation is D-separated. Therefore, the path is a causal chain, and it is observed that tampering and evacuation are conditionally independent.

3. Tampering $\perp$ Evacuation | Smoke

Path: Tampering $\rightarrow$ Alarm $\rightarrow$ Evacuation

Conditioning: Smoke (which is not on the path)

The path between Tampering and Evacuation is a causal chain through Alarm and Smoke is not a node on the path from Tampering and Evacuation, and it does not block it.

The causal path between Tampering and Evacuation is unblocked; hence, they are not D separated.

4. Tampering $\perp$ Fire

Path: None

Conditioning: None

As there is no path between tampering and fire, they are completely disconnected from each other in the Bayesian Network. As they have no path, they are D-Separated by default.

5. Tampering $\perp$ with Fire | Alarm

Path: Tampering $\rightarrow$ Alarm $\leftarrow$ Fire

Conditioning: Alarm is observed

The path between the Tampering Alarm and the Fire makes a common effect structure, and this path is blocked until the alarm is observed. Since we are conditioning on the Alarm, the path becomes unblocked, and Tampering and Fire become connected. Therefore, conditioning on the Alarm makes the nodes Tampering and Fire dependent, not independent.

6. Alarm $\perp$ Smoke

Path: Alarm $\leftarrow$ Fire $\rightarrow$ Smoke

Conditioning: None

Here, fire is a shared cause for Alarm and Smoke, and as there is no conditioning in this path, the path remains inactive unless the common parent is observed. Therefore, the path is not D-separable and is unblocked until no evidence is present.

7. Smoke $\perp$ Report

Path: Smoke $\leftarrow$ Fire $\rightarrow$ Alarm $\rightarrow$ Evacuation $\rightarrow$ Report

Conditioning: None

This is a causal chain which links Smoke to Report through several nodes in between, and as there is no conditioning along the path, the path remains active, and hence the variables are connected and are dependent. The path remains unblocked and they are not D-separated.

8. Smoke $\perp$ Tampering

Path: Smoke $\leftarrow$ Fire $\rightarrow$ Alarm $\leftarrow$ Tampering

Conditioning: None

Smoke is an effect of fire, and Alarm is a common effect of Fire and Tampering, as the common effect is not observed, the path is blocked. Thus smoke and Tampering are D-separated and are conditionally independent.

9. Smoke $\perp$ Tampering | Alarm

Path: Smoke $\leftarrow$ Fire $\rightarrow$ Alarm $\leftarrow$ Tampering

Conditioning: Alarm is observed

Now the common effect, which is alarm, is being observed, and the path is now unblocked, which shows influence between Tampering and Smoke, hence they would now be called connected and are dependent.

10. Smoke $\perp$ Tampering | Report

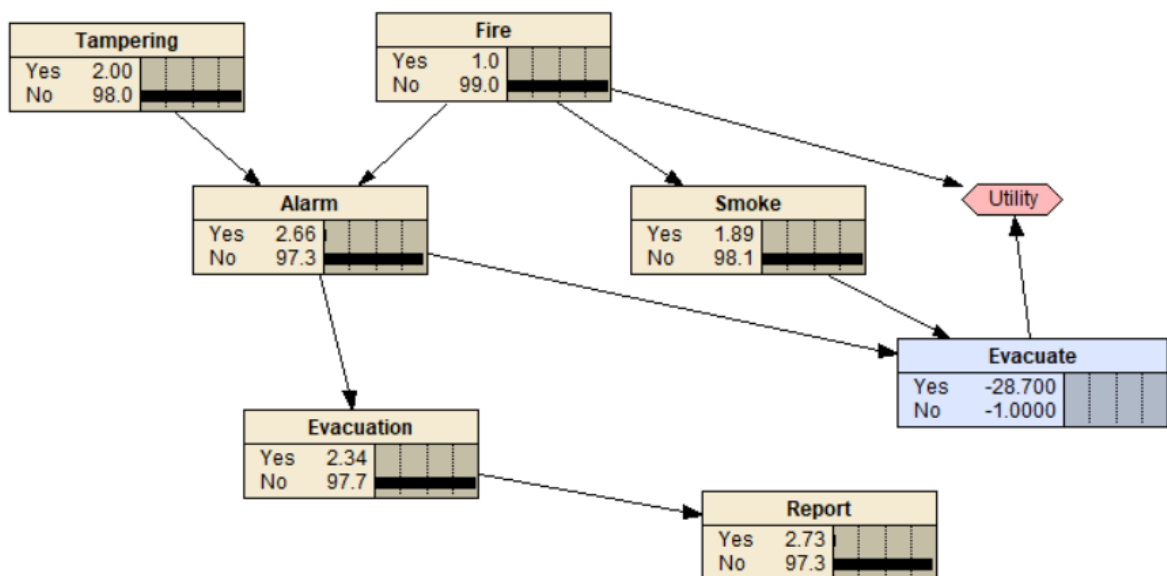
Path: Smoke $\leftarrow$ Fire $\rightarrow$ Alarm $\leftarrow$ Tampering $\rightarrow$ Evac $\rightarrow$ Report

Conditioning: Report is observed

As an alarm is a common effect of fire and tampering, making a condition on its descendant node that reports will also influence the variables and unblock the effect path. They are not D-separated.

(d) Based on your BN, construct a Bayesian Decision Network (BDN) that decides whether the building should be evacuated. Specify and justify the information links and the values in the utility node.

To support the decision of whether the building should be evacuated or not, I extended the Smoke Alarm Bayesian network by incorporating a Bayesian Decision Network structure. I added a Decision node called "Evacuate" with binary states of Yes or No in regards to the evacuation of the building. This node then receives information links from the observable variables of Alarm and Smoke. I chose these nodes as they are the key nodes in deciding whether an evacuation should happen or not, and they are observable before making a decision. Then I added a utility node to show the consequence of each of the decisions, The utility node was linked from both Evacuate and Fire as the state of Fire will show the need for evacuation.



The utility value was chosen based on the priority of keeping the residents of the building safe, and hence, the values were decided.

Fire	Evacuate	Utility
Yes	Yes	100
Yes	No	-100
No	Yes	-30
No	No	0

- Evacuation during a real fire (Fire = Yes, Evacuate = Yes) rewards the decision and gives a value of +100
- No Evacuation during a real Fire (Fire = Yes, Evacuate = No) has a strong penalty on the decision, which is of -100
- Evacuating unnecessarily during a false alarm (Fire = No, Evacuate = Yes) has a moderate reward as it suggests tampering with the alarms, hence the cost of -30 was chosen.
- Taking no action when there was no fire is the optimal response and was given a neutral utility of 0.

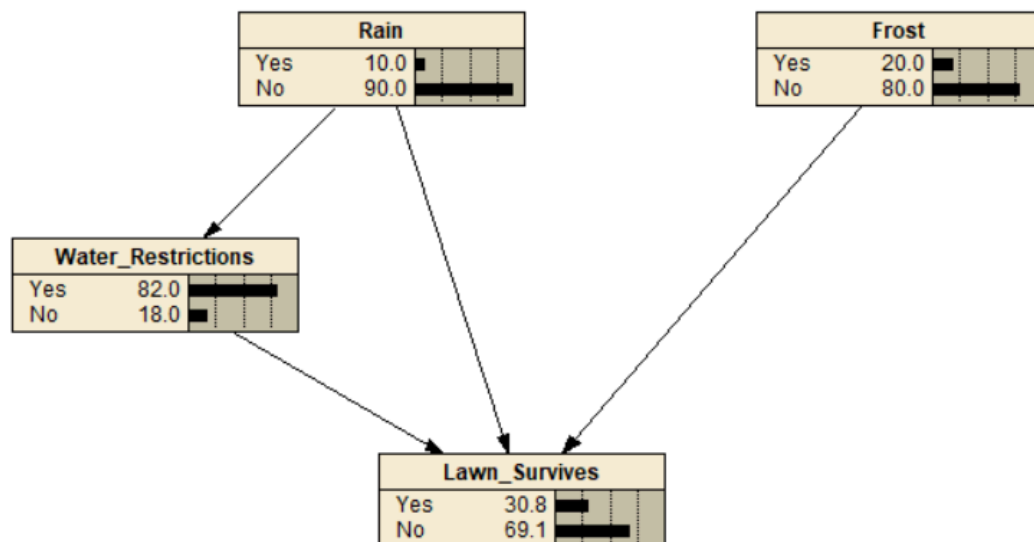
Using this configuration allows Netica to compute the expected utility of the decision options and recommend an optimal action of whether or not to evacuate the building under the different evidence scenarios.

Question 2: Bayesian Networks, Netica (16 + [2 × 2] + 13 + [1 + 3 × 2] = 40)

Winter is coming to a close, and Ron is trying to model the factors that affect the state of his lawn. After a summer of backyard cricket, the lawn is in poor shape, with several bare patches. The area has been in drought for the previous 12 months. Without rain before summer, it will be very hard to grow new grass and could cost Ron significant time and money. Furthermore, if there is no rain, the authorities could increase the level of water restrictions, meaning that Ron will be unable to water his lawn at all. This would make the chances of his lawn surviving very small indeed. To further complicate the matter, there is a small chance that the area could experience another frost before the weather warms up, which could also damage the new lawn.

(a) Design a BN accordingly. Justify your design.

Justification:



I have modelled this Bayesian Network by considering the factors affecting Ron's lawn. I have taken four key variables as stated in the question.

Variables:

- **Rain(Yes, No):** Considered as one of the most important metrics, rain affects the growth of the lawn. Based on the prolonged 12-month drought, the likelihood of rain is very low.
- **Frost(Yes, No):** Whether the area experiences another frost before the start of the summer will also affect the growth of the grass, and it can severely damage the lawn.
- **Water_Restriction(Yes, No):** This affects the regular availability of water for the lawn, and if the authorities impose a water restriction, the state of the lawn can diminish. The restriction on water depends on whether it rains or not.
- **Lawn_Survives(Yes, No):** Whether Ron's lawn survives the season and regrows successfully. This is influenced by all the other variables.

Dependencies and Linking:

1. Rain $\rightarrow$ Water_Restrictions: As stated in the question, we can observe that if it rains, the authority will have enough water, but on the other hand, if there is no rain the restriction of water restriction by the authority can be imposed and hence the water restriction depends on whether it rains or not.
2. Rain $\rightarrow$ Lawn_Survives: As rain provides natural irrigation to the lawn, these two are connected to show that proper rain will increase the likelihood of the lawn surviving.
3. Water_restriction $\rightarrow$ Lawn_Survives: Restriction of water by the authority will hinder Ron from providing artificial watering to the lawn and it will impact the growth of the lawn.
4. Frost $\rightarrow$ Lawn_Survives: A frost will directly associate with the lawn survivability as it will cause the soil to freeze up and can also kill the growth of several plants.

(b) Inspect your BN and report on any value assignments that will cause d-separation between any sets of nodes. Explain why this is the case using concepts learnt in the lecture.

D-separation helps us determine whether two variables in a Bayesian Network are conditionally independent, given a set of other variables. Below, we evaluate key variable pairs from the BN, using concepts of **common cause**, **common effect**, and **causal chain** structures.

Using the BN structure, we assess conditional independence through D-separation:

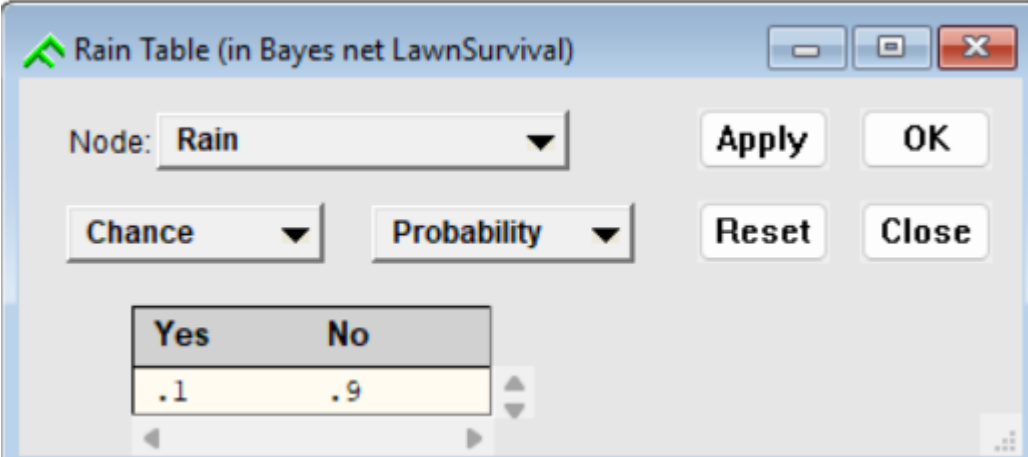
1. **Rain $\perp$ Frost: True** — These nodes are completely unconnected in the network. Since there is no path between them, they are D-separated by default and therefore conditionally independent.
2. **Frost $\perp$ Water_Restrictions | Rain: True** — Even when conditioning on Rain, Frost and Water_Restrictions remain unconnected. There is no direct or indirect active path between them, so they remain D-separated.
3. **Rain $\perp$ Lawn_Survives | Water_Restrictions: False** — Rain has a direct edge to LawnSurvives. Conditioning on Water_Restrictions does not block this path, so Rain still provides new information about Lawn_Survives, and the variables are not conditionally independent.
4. **Rain $\perp$ Lawn_Survives | (Water_Restrictions, Frost): False** — Even with both Frost and Water_Restrictions conditioned upon, the direct edge from Rain to LawnSurvives keeps the path active. Thus, Rain and LawnSurvives are not D-separated.
5. **Frost $\perp$ Lawn_Survives: False** — Frost is a direct parent of Lawn_Survives. Any change in Frost directly influences the probability of survival, meaning there is a strong dependence and no D-separation.
6. **Rain $\perp$ Water_Restrictions: False** — Rain is a direct cause of Water_Restrictions in the model structure. Knowing the value of Rain directly alters our belief about Water_Restrictions, which makes them dependent.
7. **Water_Restrictions $\perp$ Lawn_Survives | Rain: False** — Even after conditioning on Rain, Water_Restrictions still directly influences Lawn_Survives through its edge. Thus, these two variables are not D-separated.

8. **Frost $\perp$ Water_Restrictions: True** — There is no path, direct or indirect, between Frost and Water_Restrictions in the network. They are disconnected and hence conditionally independent under all circumstances.

(c) Quantify the relationships in the network by adding plausible numbers for the CPTs that match the description of the problem. Justify the numbers in your CPTs.

CPTs with justification:

1. Rain:

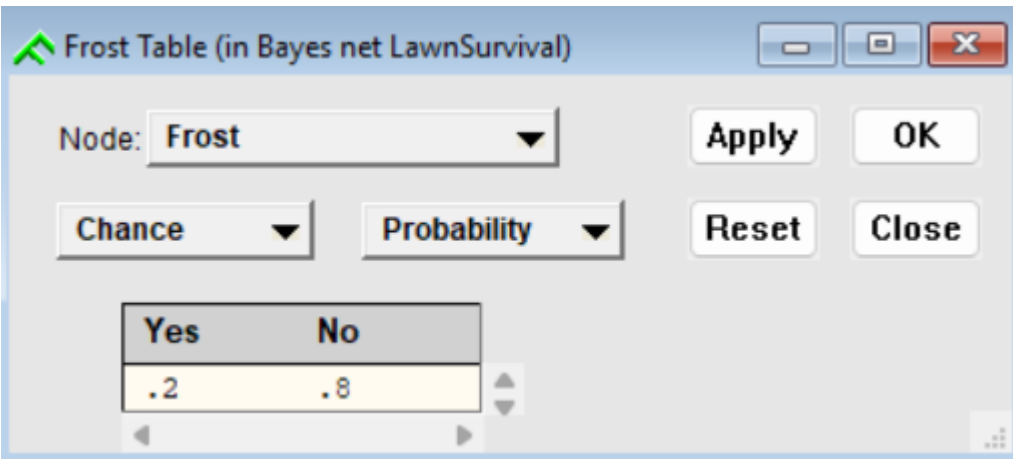


Yes	No
.1	.9

- $P(\text{Rain} = \text{Yes}) = 0.1$
- $P(\text{Rain} = \text{No}) = 0.9$

As there have been severe drought conditions over the past 12 months, this aligns with the likelihood that there might not be rain in the area.

2. Frost:



Yes	No
.2	.8

- $P(\text{Frost} = \text{Yes}) = 0.2$
- $P(\text{Frost} = \text{No}) = 0.8$

As the question suggests that there is just a small chance that there might be frost before the warming up of the season this probability lines up with that.

3. Water_Restriction:

Rain	Yes	No
Yes	.1	.9
No	.9	.1

Rain	Restrictions = Yes	No	Explanation
Yes	0.1	0.9	If there is rain, the likelihood of restrictions by the authority is less, hence the values.
No	0.9	0.1	If there is no rain in the area, the likelihood of water being restricted by the authority increases.

4. Lawn_Survives:

Lawn_Survives Table (in Bayes net LawnSurvival)

Node: **Lawn_Survives** Apply OK

Chance Probability Reset Close

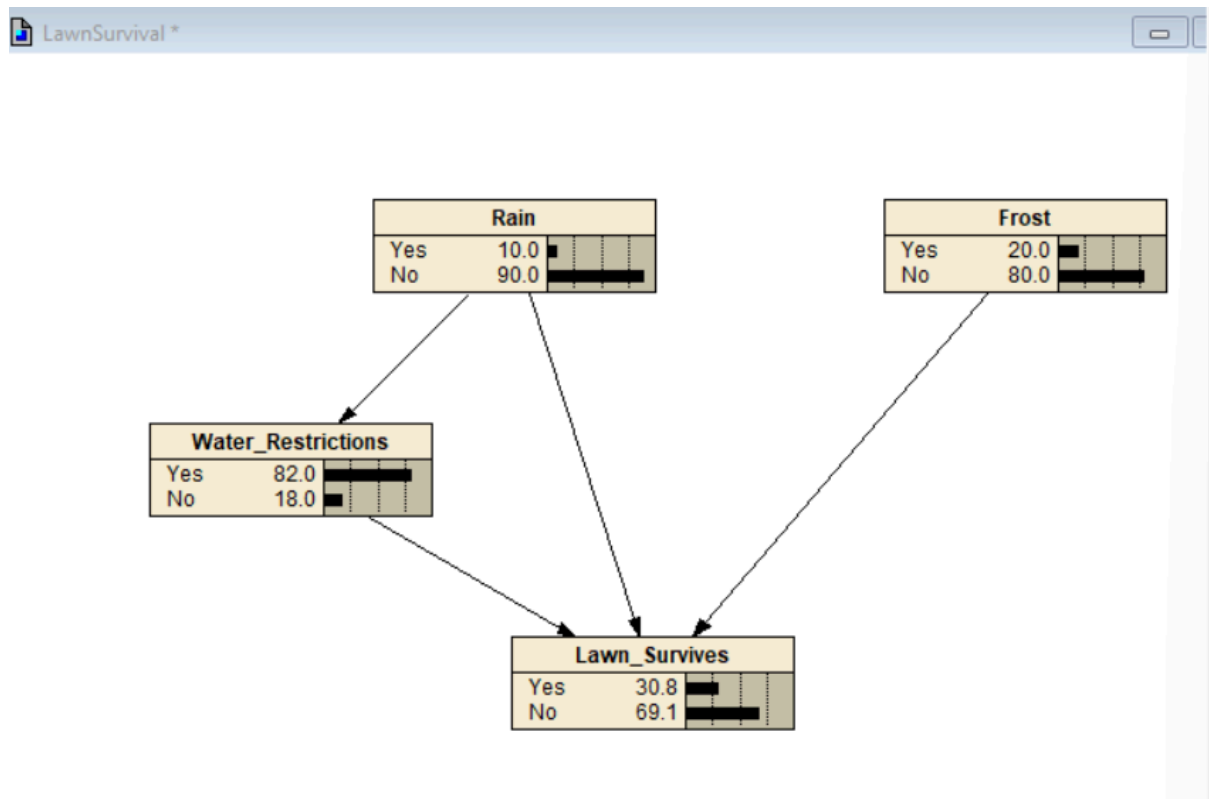
Water_Restrictions	Rain	Frost	Yes	No
Yes	Yes	Yes	.59	.41
Yes	Yes	No	.73	.27
Yes	No	Yes	.17	.83
Yes	No	No	.24	.76
No	Yes	Yes	.72	.28
No	Yes	No	.92	.08
No	No	Yes	.34	.66
No	No	No	.46	.54

Water_Restrictions	Rain	Frost	Lawn_Survives = Yes	No	Justification
Yes	Yes	Yes	0.59	0.41	As there is rain, it helps the growth of the lawn, but due to frost and water restriction, the overall survivability of the lawn is not very high.
Yes	Yes	No	0.73	0.27	Due to rain, the growth is helped, which is aided by no frost, but the unavailability of water does not ensure a high probability of survivability.
Yes	No	Yes	0.17	0.83	Without rain and water, and also the setting of frost, the likelihood of lawn survival is very low.
Yes	No	No	0.24	0.76	Without rain and water, even if frost does not happen, the survivability does not increase much.

No	Yes	Yes	0.72	0.28	If there is rain and even water restrictions are not there, then the lawn can recover from the frost, and the survival rate is high
No	Yes	No	0.92	0.08	Under perfect conditions of Rain, water availability and no frost setting, i n the survival of the lawn is almost ensured, although other realistic factors like soil condition, pests, etc., can also affect the case.
No	No	Yes	0.34	0.66	If there is no rain, the availability of water can help offset the impact of frost, but still, the survival of the lawn is low.
No	No	No	0.46	0.54	No rain, but watering and the absence of frost help balance.

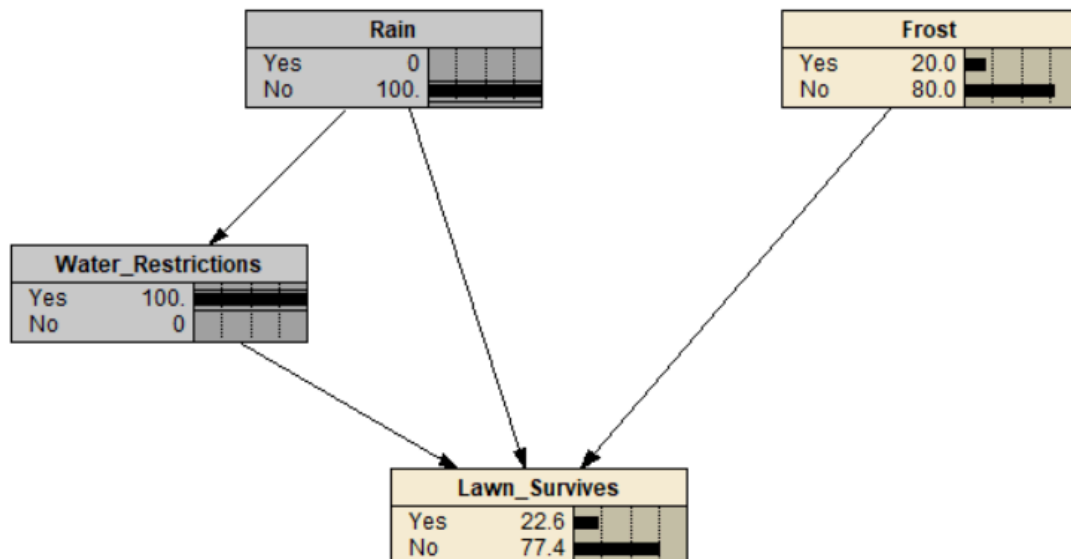
(d) Using Netica, demonstrate the workings of your BN by determining the probability of the lawn growing in the following cases.

1. There is no evidence.



If there is no evidence in the situation, then we see that the chance for rain is low at 10%, and the chance of frost happening is also less at 20%. Computing from the following values, we see that the chance of water being restricted is high at 82%. The survival of the lawn in such a situation is only 30.8%. This is in line with our observations that under the low possibility of rain and a high chance of water being restricted, even without frost, the lawn's survival is affected negatively.

2. There is no rain, and water restrictions have been applied. Explain your results compared to the previous case.



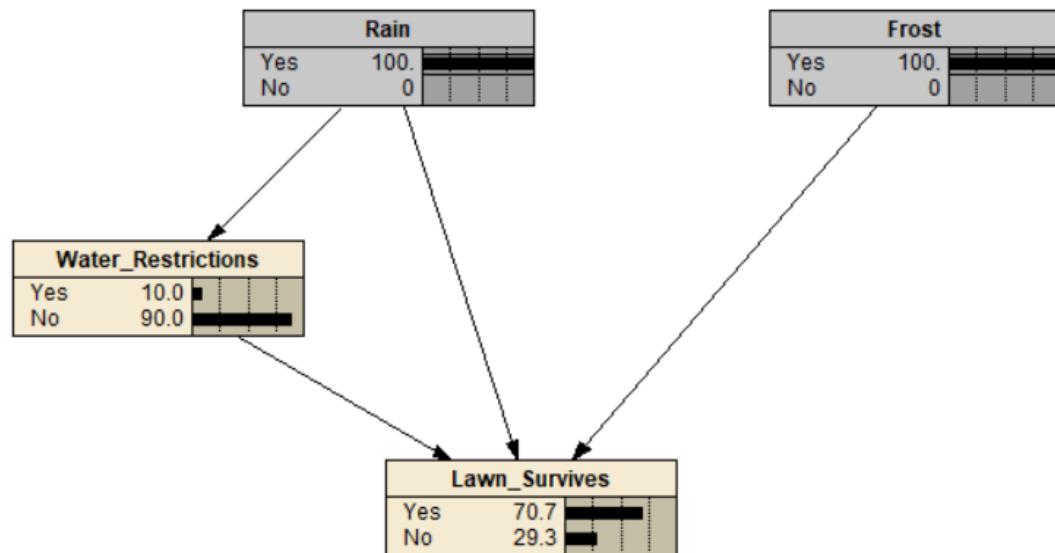
Under the circumstances where:

- Rain = No
- Water_Restriction = Yes

This shows a damaging condition for the lawn, where there is no source of irrigation. Here, the survivability of the lawn depends on frost now, which has a 20% chance of occurring (default value).

The survivability of the lawn drops from 30.8% in the previous no-evidence case to 22.6% here when water is not available.

3. There is frost, but it has rained. Explain your results compared to the previous 2 cases.



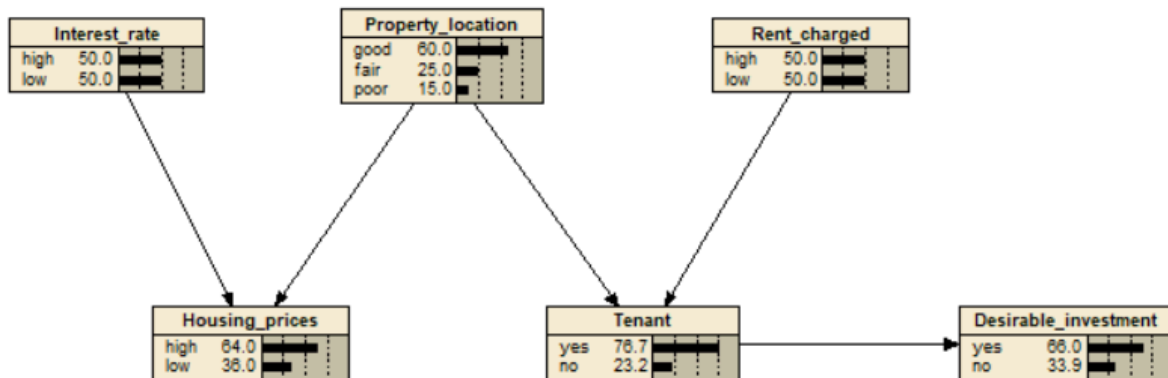
Under the circumstances where:

- Rain = Yes
- Frost = Yes

Since surely that rain will happen, the chance of water restriction comes down to 10%. But as there is a certainty of frost, we see the Lawn Survival probability only go up to 70.7%. This suggests that even if there is ample rain and water restriction is not there, the survival of the lawn will be affected by the certainty of frost. Compared to the last 2 observations, we see that the availability of water highly impacts the survival of the lawn and brings up the probability to 70.7%.

Question 3 (9 + 5 + 6 = 20 marks) Consider the following Bayesian Network called rental2.dne (available on Moodle).

(a) List the conditions under which you can propagate evidence from the Interest rate node to the Rent charged node. Which nodes would need to be instantiated or uninstantiated? Explain why this is the case using concepts learnt in the lecture.



By observing the path from Interest_rate to Rent_charged, we see that:

Interest_rate → Housing_prices ← Property_location → Tenant ← Rent_charged

Here we see that there are two colliders in the path:

- For: Interest_rate → Housing_prices ← Property_location: Housing_prices is a collider between Interest_rate and Property_location
- For: Property_location → Tenant ← Rent_charged: Tenant is a collider between Property_location and Rent_charged

From the D-separation rule, we know that, in a v-structure like $A \rightarrow C \leftarrow B$, the path is blocked unless C (Housing_prices, Tenant) or its descendant is observed.

Therefore, from the observation, we can infer that evidence from Interest_rate can propagate to the Rent_charged node only if:

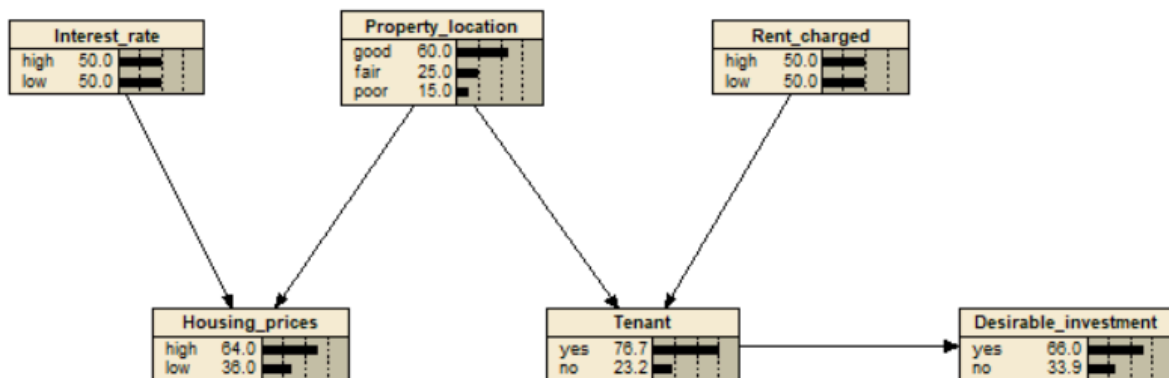
- Housing_prices is observed
- Tenant is also observed
- And Property_location should be unobserved

As the only path between Interest_rate and Rent_charged is:

Interest_rate → Housing_prices ← Property_location → Tenant ← Rent_charged.

This path is blocked at two colliders: Housing_prices and Tenant, and evidence can only travel if both these nodes are instantiated, and property_location should be uninstantiated.

(b) Repeat question (a) for propagating evidence from Desirable investment to Housing prices. Explain why this is the case using concepts learnt in the lecture.



Now observing the path from Desirable_investment to Housing_prices, we see that:

$\text{Desirable_investment} \leftarrow \text{Tenant} \leftarrow \text{Property_location} \rightarrow \text{Housing_prices}$

This forms a reverse causal chain and in a reverse causal chain the path is active only when all the intermediate nodes are unobserved.

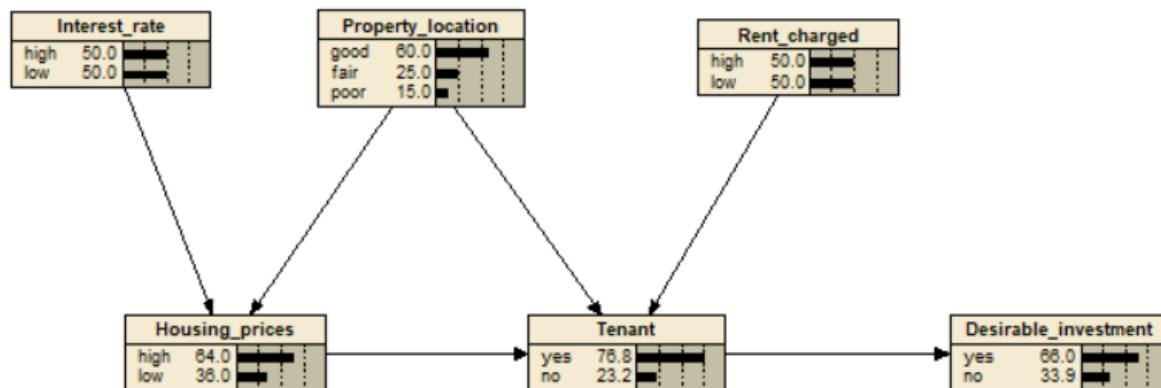
Here we can see that there are two intermediate nodes:

- Tenant
- Property_location

Therefore, in this reverse causal chain according to D - separation rule, evidence can only flow from Desirable_investment to Housing_prices if the intermediate nodes i.e. Tenant and Property_location remains unobserved. The path will be blocked if either of them are observed.

(c) Repeat the above questions under the assumption that there is also an arc from Housing prices to Tenant (the corresponding BN, rental3.dne, is available on Moodle). Provide explanations using concepts learnt in the lecture.

Justification:



(c.1) List the conditions under which you can propagate evidence from the Interest rate node to the Rent charged node. Which nodes would need to be instantiated or uninstantiated? Explain why this is the case using concepts learnt in the lecture.

Now we see the possible path from Interest_rate to Rent_charged is:

Interest_rate → Housing_prices → Tenant ← Rent_charged

This is a V-structure with the Tenant node having two parent nodes (Housing_prices and Rent_charged). According to the D-separation rule we know that in a V-structure the path remains blocked until the collider or its descendent is observed.

Hence, we can say that Evidence from Interest_rate can propagate to Rent_charged only if the Tenant is observed.

(c.2) Repeat question (a) for propagating evidence from Desirable investment to Housing prices. Explain why this is the case using concepts learnt in the lecture.

Here, the path from Desirable investment to Housing prices is:

Desirable_investment ← Tenant ← Housing_prices

This forms a reverse causal chain from Housing_prices to Desirable_investment.

From the D-Separation rule, we know that in a causal chain or reverse chain, the path will remain active unless an intermediate node is observed. Here, as the Tenant node is an intermediate node, we can say that evidence from Desirable_investment can propagate to Housing_prices only if Tenant is unobserved.